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SCHOOL OF ELECTRICAL AND ELECTRONICS ENGINEERING

ELECTRONICS AND COMMUNICATION PROGRAMME

&

IEEE Student Branch STB30191

Dr. Ambedkar Institute of Technology

PRESENTS

CIRCUIT RUSH

QUIZ ✦ DEBUG ✦ BUILD

RULE BOOK

DATE :

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VENUE :

ECE Seminar

Hall & Lab

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1. EVENT INTRODUCTION

1.1 Event Name : - "CIRCUIT RUSH – A Technical Challenge**"**

1.2 Theme:- Think. Design. Debug. Innovate.

Circuit Rush is based on the theme of practical electronics, circuit analysis and technological innovation. It provides participants with an opportunity to demonstrate their technical knowledge, problem-solving abilities and hardware implementation skills through a series of challenging rounds.

1.3 Objectives

The main objectives of Circuit Rush are:

- To test participants' fundamental knowledge of electronics and emerging technologies.
- To develop analytical thinking and circuit-debugging skills.
- To encourage practical learning through hands-on hardware implementation.
- To enhance creativity and innovation in designing IoT-based solutions.
- To improve teamwork, time management and problem-solving abilities.
- To bridge the gap between theoretical knowledge and practical engineering applications.

1.4 Event Overview

Circuit Rush is a three-round technical event designed to evaluate participants' theoretical knowledge, practical circuit skills and innovative thinking.
The event consists of the following rounds:

Round 1: Speed-Based Technical Quiz

An app-based technical quiz covering five domains: Basic Electronics and Device Physics, Digital Electronics, IoT, VLSI and one additional domain to be announced. Teams must answer questions within a limited time.

Round 2: Analog Circuit Debugging

A hands-on hardware challenge focused on Operational Amplifiers (Op-Amps) and 555 Timer configurations. Participants must identify the required circuit, draw its schematic and successfully implement it using hardware components

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Round 3: IoT Hardware Mini-Hackathon

A three-hour hardware development challenge in which teams design and implement IoT-based solutions using microcontrollers and IoT peripherals to solve the given problem statements.
Each round evaluates different technical abilities, and teams progress based on their performance and the specified qualification criteria.

2. EVENT STRUCTURE

2.1 Competition Format

Circuit Rush is a three-round technical competition designed to test participants' theoretical knowledge, practical circuit-solving abilities and hardware development skills.

The event will be conducted in three consecutive rounds, with teams qualifying for the next round based on their performance in the previous round.

2.2 Round Structure

Round 1: Speed-Based Technical Quiz

- Mode: App-based digital quiz.
- Total questions: 25.
- Domains: 5 technical domains.
- Time limit: 80 seconds per question.
- Maximum score: 250 points.
- Qualification: Top 25 teams advance to Round 2.

Round 2: Analog Circuit Debugging

- Mode: Hands-on hardware debugging.
- Focus areas: Op-Amp and 555 Timer configurations.
- Prepared circuits: 10 circuits of identical difficulty.
- Execution: Three staggered batches.
- Maximum score: 250 points.
- Qualification: Top 10 teams advance to Round 3.

Round 3: IoT Hardware Mini-Hackathon

- Mode: Hardware development challenge.
- Duration: 3 hours.
- Resources: Microcontrollers and IoT peripherals.
- Phases: Documentation and hardware execution.
- Maximum score: 500 points.
- Evaluation: Based on documentation quality and prototype functionality.

2.3 Scoring Distribution

Round Maximum marks

- Round 1 250
- Round 2 250
- Round 3 500
- Total 1,000

Scoring distribution

- Round 1 25%
- Round 2 25%
- Round 3 50%



2.4 Team Progression

The Qualification process will be as Follows:

2.5 General Competition Guidelines

- All teams must participate in the rounds in the prescribed order.
- Only qualified teams will be permitted to participate in the subsequent rounds.
- Participants must complete each round within the specified time limit.
- Scores will be recorded and verified by the respective evaluators.
- Any unfair practices may result in penalties or disqualification.
- The organising committee reserves the right to resolve unforeseen situations during the competition.

3.RULES AND REGULATIONS

All participants are required to follow the rules and regulations mentioned below to ensure fair competition, discipline and smooth execution of the event.

3.1 General Participation Rules

- All participants must complete the registration process before participating in the event.
- Participants must report to the venue at the specified reporting time.
- Teams must maintain the same members throughout all three rounds.
- Participants must carry their valid college ID cards.
- Every team must follow the instructions provided by the event coordinators and organising committee.
- Participants must maintain discipline and professional behaviour throughout the event.

3.2 Competiton Rules

- The competition will consist of three rounds conducted in the prescribed order.
- Only teams qualifying in the previous round will be allowed to participate in the next round.
- Participants must complete each task within the allotted time.
- Teams must work independently and must not exchange answers, circuit diagrams, code or solutions with other teams.
- Only authorised devices, components and resources may be used during the competition.
- The decisions of the judges regarding technical evaluation and scoring will be final.

3.3 Technical And Equipment Guidelines

- Participants must handle all electronic components, instruments and equipment carefully.
- Any damaged or malfunctioning component must be immediately reported to the technical coordinators.
- Teams must not modify or misuse the hardware resources provided by the organisers.
- Proper safety precautions must be followed while working with electrical and electronic circuits.
- Participants must disconnect the power supply before modifying circuit connections, unless instructed otherwise by the evaluator.
- All components and equipment provided by the organisers must be returned after completion of the respective round

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3.4 Fair Play and Ethical Conduct

- Any form of cheating, plagiarism or unauthorised assistance is strictly prohibited.
- Participants must not interfere with the work or equipment of other teams.
- Sharing solutions or obtaining answers through external sources during the competition is not permitted unless explicitly authorised.
- Any attempt to manipulate scores or influence the judging process may lead to disqualification.
- Participants must respect fellow competitors, judges, volunteers and organising committee members.

3.5 Time Management

- Each round will have a specified time limit.
- Participants must complete and submit their tasks within the allotted duration.
- Late submissions will not be considered unless an exception is approved by the organisers.
- Teams must stop their work immediately when the time limit is announced.
- No additional time will be provided unless approved by the organising committee.

3.6 Disqualification

- A team may be disqualified under the following circumstances:
- Involvement in cheating or unfair practices.
- Use of unauthorised external assistance.
- Deliberate damage or misuse of event equipment.
- Misconduct or repeated violation of event instructions.
- Interference with other teams or disruption of the competition.

3.7 Organising Committee Authority

- The organising committee reserves the right to make necessary decisions regarding event management.
- Any unforeseen issues will be addressed by the organising committee.
- Changes to the event schedule or rules, if necessary, will be communicated to participants.
- The organising committee's decisions regarding event conduct and rule violations will be final.

4. PARTICIPATION GUIDELINES

All participants must follow the guidelines mentioned below to ensure smooth registration, participation and coordination throughout the event.

4.1 Eligibility

- The event is open to eligible students as specified by the organising committee.
- Participants must provide valid college identification during registration or verification.
- All participants must complete the registration process before the deadline.

4.2 Team Formation

- Participants must register as teams according to the team-size limit specified by the organisers.
- Each team must have a designated team leader to coordinate with event volunteers.
- Team members must remain unchanged throughout all three rounds unless approved by the organising committee.

4.3 Registration Guidelines

- Teams must submit accurate details of all team members during registration.
- Incomplete or incorrect registration details may lead to rejection of the registration.
- Registration confirmation will be provided by the organising committee.
- Any registration-related queries must be addressed to the event coordinators.

4.4 Participant responsibilities

- Participants must report to the venue at the specified time.
- Teams must carry their college ID cards.
- Participants must bring only the permitted materials and equipment.
- Every team must maintain discipline and cooperate with the organisers.
- Participants must follow all technical and safety instructions during hardware activities.

4.5 Attendance and Reporting

- All registered teams must be present before the commencement of each round.
- Late reporting may result in restricted participation, subject to the organisers' decision.
- Participants must remain available until their respective round is completed.

5. EVALUATION CRITERIA

The performance of each team will be evaluated based on their technical knowledge, problem-solving abilities, practical implementation and innovation. Each round carries a specific maximum score, as mentioned below.

5.1 Round 1 – Speed-Based Technical Quiz (250 Points)

- Each correct answer carries 10 points.
- A total of 25 questions will be conducted across five technical domains.
- Each question must be answered within 80 seconds. The maximum score is 250 points.

5.2 Round 2 – Analog Circuit Debugging (250 Points)

- Independent Deduction – 100 Points: Awarded for correctly identifying the required circuit and drawing its schematic without assistance.
- Hardware Execution – 150 Points: Awarded for successfully constructing and solving the physical circuit.
- Hint Penalty: Teams requesting a hint will forfeit the 100 independent-deduction points and can earn a maximum of 150 points.

5.3 Round 3 – IoT Hardware Mini-Hackathon (500 Points)

Documentation – 100 Points: Awarded for submitting the proposed solution according to the prescribed format.

- Tier A – 400 Points: Prototype is fully functional before the teacher review phase.
- Execution Evaluation:
- Tier B – 300 Points: Prototype becomes functional after teacher intervention.
- Tier C – 100 Points: Non-working prototype demonstrating a valid attempt.
- Note: Round 3 execution scores are awarded according to one applicable tier only. The maximum score for the round is 500 points.

6.DISQUALIFICATION AND PENALTIES

All participants are expected to follow the event rules and maintain fair competition. Any violation of the rules may result in penalties or disqualification depending on the seriousness of the offence.

6.1 Grounds for Disqualification

- A team may be disqualified under the following circumstances:
- Involvement in cheating, plagiarism or any unfair practices.
- Using unauthorised external assistance during the competition.
- Deliberately damaging or misusing the hardware components and equipment.
- Interfering with the work or performance of other teams.
- Providing false registration details or violating participation guidelines.

6.2 Penalties for Rule Violations

- Teams violating competition rules may receive warnings or score deductions.
- Misuse of equipment or failure to follow safety instructions may result in removal from the respective round.
- Repeated misconduct or serious violations may lead to immediate disqualification.
- Any attempt to manipulate scores or influence the evaluation process will be treated as a serious offence.

6.3 Disqualification Procedure

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- Any reported violation will be reviewed by the organising committee.
- The concerned team may be asked to provide an explanation regarding the violation.
- The organising committee will decide the appropriate penalty based on the severity of the violation.
- The decision of the organising committee regarding disciplinary matters will be final.
- Important Note: All participants must maintain discipline, integrity and sportsmanship throughout the event. The organising committee reserves the right to take necessary action against any violation.

7. GENERAL INSTRUCTIONS

All participants are expected to follow the general instructions to ensure the smooth, safe and successful conduct of Circuit Rush.

7.1 Event Discipline and Conduct

- Participants must maintain proper discipline and decorum throughout the event.
- All teams must follow the instructions provided by the event coordinators and volunteers.
- Participants must respect judges, organisers, volunteers and fellow competitors.
- Any disputes or queries must be addressed politely to the event coordinators.
- Unnecessary disturbance or disruptive behaviour during the competition is strictly prohibted.

7.2 Safety and Equipment Handling

- Participants must handle all electronic components, tools and equipment with proper care.
- Electrical connections must be checked before switching on the power supply.
- Any damaged or malfunctioning equipment must be immediately reported to the technical team.
- Participants must keep their workstations clean and organised.
- All borrowed components and equipment must be returned to the organisers after the completion of the event.

7.3 Organizing Committe Instructions

- Participants must strictly follow the event schedule and instructions announced by the organising committee.
- Any changes in the event schedule, rules or procedures will be communicated to the participants.
- The organising committee reserves the right to address unforeseen technical or organisational issues.
- Any situation not covered in this rulebook will be resolved by the organising committee.
- The decisions of the organising committee regarding event management and conduct will be final.
- Important Note: All participants are expected to cooperate with the organising committee and contribute towards maintaining a fair, safe and technically engaging competition.

8.PRIZE POOL AND PARTICIPATION CERTIFICATES

8.1 Prize Distribution

The top three winning teams will be rewarded with cash prizes in recognition of their technical skills, innovation, teamwork and performance throughout the event.

8.1 Position Cash Prize

- 1st Prize ₹5,000
- 2nd Prize ₹3,000
- 3rd Prize ₹2,000
- Total Prize Pool ₹10,000

8.2 Participation Certificates

- Participation certificates will be provided to all registered participants who successfully take part in the event.
- Certificates will recognise the participants' involvement and enthusiasm in the technical competition.
- Winners will receive recognition for securing the top three positions.

8.3 Awards and Recognition

- The top three teams will be recognised based on their overall performance and evaluation.
- Prize distribution will take place during the valedictory or closing ceremony.
- The organising committee's decision regarding final results and prize distribution will be considered final.